

6

NUMBER AND ALGEBRA

EQUATIONS

In 1962, when astronaut John Glenn was preparing to be the first American to orbit the Earth, he called upon mathematician Katherine Johnson (1918–2020) to check by hand the complex calculations being performed on NASA's new electronic computers. Johnson worked for NASA for 35 years and her story was told in the 2016 book and film *Hidden Figures*. During her career, she solved equations that guided the paths of spacecrafts and helped the US send the first astronauts to the moon in 1969. Building on the ancient achievements of the Egyptians and Babylonians, Johnson opened the door to space travel and future space tourism.



Chapter outline

		Working mathematically				
6.01	Equations	U	F			
6.02	Equations with algebraic fractions*	U	F			
6.03	Quadratic equations $x^2 + bx + c = 0$*	U	F		R	C
6.04	Equation problems	U	F	PS	R	C
6.05	Equations and formulas	U	F		R	C
6.06	Changing the subject of a formula*	U	F		R	C

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Wordbank

equation A mathematical statement that 2 quantities are equal, involving algebraic expressions and an equals sign (=)

formula A rule written as an algebraic equation, using variables

quadratic equation An equation involving a variable squared (power of 2), such as $3x^2 - 6 = 69$

solution The answer to an equation or problem; the correct value(s) of the variable that makes an equation true

subject of a formula The variable for which a formula is written, on the left side of the formula, for example, for $E = mc^2$, the subject is E

Quiz
Wordbank 6

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communication

Videos (10):

6.01 Equations with variables on both sides • Equations with brackets

6.02 Equations with algebraic fractions • Solving equations with fractions

6.03 Simple quadratic equations • Quadratic equations by factorising

6.05 Equations and formulas 2 • Equations and formulas 1 • Formulas and equations

6.06 Changing the subject of a formula

Quizzes (5):

- Wordbank 6
- SkillCheck 6
- Mental skills 6
- Language of maths 6
- Test yourself 6

Worksheets (2):

6.06 Changing the subject of a formula

Mind map Equations

Puzzles (9):

6.01 Equations with unknowns on both sides • Equations

6.02 Equations code puzzle • Equations order activity • Solving linear equations 1 • Solving linear equations 2

6.05 Getting the right formula

6.06 Formulas squaresaw

Language of maths Equations crossword

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

- solve linear equations and problems involving equations
- (PATH) solve linear equations involving simple algebraic fractions
- solve quadratic equations of the form $ax^2 = c$ and (PATH) $x^2 + bx + c = 0$
- use formulas to solve problems
- (PATH) change the subject of a formula

Quiz
SkillCheck 6

SkillCheck ANSWERS ON P. 595

1 Solve each equation.

a $8y = 16$ **b** $10x = 120$ **c** $\frac{m}{5} = 2$

d $w + 6 = 10$ **e** $m - 3 = 12$ **f** $n + 6 = -4$

2 Expand each expression.

a $5(x + 10)$ **b** $4(y - 1)$ **c** $2(5 - 3y)$

3 Solve each equation.

a $2x + 3 = 23$ **b** $3x - 5 = 19$ **c** $4a + 5 = 2a - 19$

d $\frac{3x+2}{5} = 4$ **e** $4(2 - x) = -24$

Equations

6.01

Example 1

Solve each equation.

a $3m - 6 = 12$

b $5 - 2a = 3a$

c $9x + 10 = 7x - 6$

d $5(p + 6) = 3p + 5$

Solution

a $3m - 6 = 12$

$3m - 6 + 6 = 12 + 6$ Adding 6 to both sides

$3m = 18$

$\frac{3m}{3} = \frac{18}{3}$ Dividing both sides by 3

$m = 6$

Check:

LHS $= 3 \times 6 - 6 = 12$

RHS $= 12$

LHS = RHS

b $5 - 2a = 3a$

$5 - 2a + 2a = 3a + 2a$ Adding $2a$ to both sides

$5 = 5a$

$5a = 5$

$\frac{5a}{5} = \frac{5}{5}$ Dividing both sides by 5

$a = 1$

Check:

LHS $= 5 - 2 \times 1 = 3$

RHS $= 3 \times 1 = 3$

LHS = RHS

c $9x + 10 = 7x - 6$

$9x + 10 - 7x = 7x - 6 - 7x$

$2x + 10 = -6$

$2x + 10 - 10 = -6 - 10$

$2x = -16$

$\frac{2x}{2} = \frac{-16}{2}$

$x = -8$

d Expand the LHS:

$5(p + 6) = 3p + 5$

$5p + 30 = 3p + 5$

$5p + 30 - 3p = 3p + 5 - 3p$

$2p + 30 = 5$

$2p + 30 - 30 = 5 - 30$

$2p = -25$

$\frac{2p}{2} = \frac{-25}{2}$

$p = -12\frac{1}{2}$

Puzzles
Equations with unknowns on both sides

Equations

Videos
Equations with variables on both sides

Equations with brackets

Example 2

Solve each equation.

a $4(y+1)+3(y-5)=8$

b $\frac{5d}{3}+4=15$

c $\frac{6-m}{5}=4$

d $\frac{4w+3}{7}=5$

Solution

a Expand and collect like terms.

$4(y+1)+3(y-5)=8$

$4y+4+3y-15=8$

$7y-11=8$

$7y-11+11=8+11$

$7y=19$

$\frac{7y}{7}=\frac{19}{7}$

$y=2\frac{5}{7}$

Check:

$\text{LHS}=4\left(2\frac{5}{7}+1\right)+3\left(2\frac{5}{7}-5\right)=8$

$\text{RHS}=8$

$\text{LHS}=\text{RHS}$

b $\frac{5d}{3}+4=15$

$\frac{5d}{3}+4-4=15-4$

$\frac{5d}{3}=11$

$\frac{5d}{3}\times 3=11\times 3$

$5d=33$

$d=\frac{33}{5}$

$d=6\frac{3}{5}$

c $\frac{6-m}{5}=4$

$\frac{6-m}{5}\times 5=4\times 5$

$6-m=20$

$6-m-6=20-6$

$-m=14$

$m=-14$

d $\frac{4w+3}{7}=5$

$\frac{4w+3}{7}\times 7=5\times 7$

$4w+3=35$

$4w+3-3=35-3$

$4w=32$

$w=\frac{32}{4}$

$w=8$

EXERCISE 6.01 ANSWERS ON P. 595

Equations U F

EXAMPLE 1

1 Solve each equation.

a $\frac{k}{6}=10$

b $w+3=-6$

c $5y-1=9$

d $3a+10=25$

e $2x+6=22$

f $15a-2=13$

g $12-r=18$

h $7w-10=32$

i $9y-6=-24$

j $11-6a=-10$

k $7u=u+32$

l $5a=a-7$

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2 Solve $6x - 3 = 27$. Select the correct answer **A**, **B**, **C** or **D**.

A $x = 4$ **B** $x = 5$ **C** $x = 10$ **D** $x = 18$

3 Solve $10 - 2a = 20$. Select **A**, **B**, **C** or **D**.

A $a = -15$ **B** $a = 8$ **C** $a = 32$ **D** $a = -5$

4 Solve each equation.

a $5y + 10 = 3y + 30$ **b** $8a + 20 = 4a + 10$

c $6y - 1 = 3y + 14$ **d** $12a + 30 = 5a + 9$

e $5y + 3 = 8y - 21$ **f** $14x - 20 = 8x - 14$

g $9y + 1 = 3y - 5$ **h** $15x - 15 = 8x - 85$

i $8m - 10 = 5 - 2m$ **j** $18 - 3y = 6 - 2y$

k $1 - 7a = 10 + 2a$ **l** $11 - 5x = 3x + 43$

5 Solve $4y = y - 15$. Select **A**, **B**, **C** or **D**.

A $y = -3$ **B** $y = \frac{7}{4}$ **C** $y = -5$ **D** $y = 11$

6 Solve each equation.

a $3(x - 6) = 30$ **b** $5(m + 10) = 80$

c $2(5y + 3) = 46$ **d** $3(y + 2) = 5y - 10$

e $5(y + 4) = 3y + 6$ **f** $10(x - 3) = 5(x + 5)$

g $2(3m + 6) = 4(m - 1)$ **h** $5(2a + 7) = 5(4 - a)$

i $3(1 - 2y) = 18 - 3y$

7 Solve $2(y - 3) = 5 + 4y$. Select **A**, **B**, **C** or **D**.

A $y = -9$ **B** $y = -5$ **C** $y = -\frac{11}{2}$ **D** $y = -\frac{1}{2}$

8 Solve each equation.

a $3(d + 3) + 4(d + 1) = 15$ **b** $3(y - 1) + 5(y + 4) = 10$

c $7(k + 1) + 2(k - 6) = 3$ **d** $5(g - 3) + 2(g - 2) = 4$

e $6(2h + 3) + 5(h - 3) = 9$ **f** $2(1 + p) + 3(4 + p) = 5$

9 Solve each equation.

a $\frac{4k}{5} = 8$ **b** $\frac{2d}{6} = 7$ **c** $\frac{w+7}{3} = 8$

d $\frac{3m}{4} + 6 = 10$ **e** $\frac{7+a}{8} = 3$ **f** $3 + \frac{2d}{7} = 9$

g $\frac{9-y}{5} = 3$ **h** $8 - \frac{4x}{3} = 5$ **i** $10 = \frac{9b}{7}$

j $\frac{x+1}{4} + 2 = 10$ **k** $\frac{y-2}{5} - 6 = 3$ **l** $\frac{m+2}{5} - 1 = 3$

m $\frac{4h-3}{2} = 7$ **n** $\frac{2e+11}{4} = 6$ **o** $\frac{12-5y}{3} = 9$

p $\frac{8h+3}{5} = 7$ **q** $\frac{9a}{4} - 5 = 7$ **r** $\frac{9-3c}{7} = 8$

6.01

EXAMPLE 2

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Investigation

Make your own equation

Here are 2 equations that have the same solution, $x = 6$:

$5x - 1 = 23 + x$ and $\frac{3x+12}{10} = 3$

1 For each solution below, make up 2 equations that have that solution.

a $x = 4$ **b** $x = \frac{1}{2}$ **c** $x = 10$

d $x = 1.5$ **e** $x = 0$ **f** $x = -2$

2 Compare your answers with those of other students. Check that each equation is correct.

6.02 Equations with algebraic fractions

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Puzzles
Equations code puzzle

Equations order activity

Solving linear equations 1

Solving linear equations 2

Example 3

Solve each equation.

a $\frac{5k+4}{3} - 7 = 12$

b $\frac{3(p-3)}{2} + 6 = 1$

Solution

a

$$\frac{5k+4}{3} - 7 = 12$$

$$\frac{5k+4}{3} - 7 + 7 = 12 + 7$$

$$\frac{5k+4}{3} = 19$$

$$\frac{5k+4}{3} \times 3 = 19 \times 3 \quad \text{Multiply both sides by 3}$$

$$5k + 4 = 57$$

$$5k + 4 - 4 = 57 - 4$$

$$5k = 53$$

$$\frac{5k}{5} = \frac{53}{5}$$

$$k = 10\frac{3}{5}$$

b

$$\frac{3(p-3)}{2} + 6 - 6 = 1 - 6$$

$$\frac{3(p-3)}{2} = -5$$

$$\frac{3(p-3)}{2} \times 2 = -5 \times 2 \quad \text{Multiply both sides by 2}$$

$$3(p - 3) = -10$$

$$3p - 9 = -10 \quad \text{Expand the LHS}$$

$$3p - 9 + 9 = -10 + 9$$

$$3p = -1$$

$$\frac{3p}{3} = \frac{-1}{3}$$

$$p = -\frac{1}{3}$$

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For equations with more than one fraction, multiply both sides by a common multiple of the denominators to remove the fractions.

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Videos
Equations with algebraic fractions

Solving equations with fractions

6.02

Example 4

Solve each equation.

a $\frac{2m}{3}-\frac{m}{2}=2$ **b** $\frac{2a+4}{5}=\frac{2}{3}$ **c** $\frac{3x-5}{4}=\frac{x+1}{7}$

Solution

a

$$\frac{2m}{3}-\frac{m}{2}=2$$

$$6\left(\frac{2m}{3}-\frac{m}{2}\right)=6\times 2$$

Multiply both sides by 6, the lowest common multiple (LCM) of 3 and 2

$$6^{2}\left(\frac{2m}{3_{1}}\right)-6^{3}\left(\frac{m}{2_{1}}\right)=12$$

$$4m-3m=12$$

$$m=12$$

Check by substituting that this solution is correct.

Note: We can also solve this equation by simplifying the LHS first:

$$\frac{2m}{3}-\frac{m}{2}=2$$

$$\frac{4m}{6}-\frac{3m}{6}=2$$

$$\frac{m}{6}=2$$

$$m=12$$

However, the first method may be faster and easier because operations with fractions are not required.

b

$$\frac{2a+4}{5}=\frac{2}{3}$$

$$15^{3}\left(\frac{2a+4}{5_{1}}\right)=15^{5}\left(\frac{2}{3_{1}}\right)$$ LCM = 15

$$3(2a+4)=10$$

$$6a+12=10$$

$$6a=-2$$

$$a=\frac{-2}{6}$$

$$a=-\frac{1}{3}$$

c

$$\frac{3x-5}{4}=\frac{x+1}{7}$$

$$28^{7}\left(\frac{3x-5}{4^{1}}\right)=28^{4}\left(\frac{x+1}{7^{1}}\right)$$ LCM = 28

$$7(3x-5)=4(x+1)$$

$$21x-35=4x+4$$

$$17x=39$$

$$x=\frac{39}{17}$$

$$x=2\frac{5}{17}$$

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EXERCISE 6.02 ANSWERS ON P. 596

Equations with algebraic fractions U F

1 Solve each equation. Select the correct answer **A**, **B**, **C** or **D**.

a $\frac{3y}{4}=6$

A $y=9$ **B** $y=8$ **C** $y=7$ **D** $y=6$

b $\frac{a+1}{2}=3$

A $a=4$ **B** $a=5$ **C** $a=6$ **D** $a=7$

c $\frac{x-1}{4}+2=10$

A $x=49$ **B** $x=37$ **C** $x=33$ **D** $x=3$

EXAMPLE 3

2 Solve each equation.

a $\frac{2y-3}{5}+2=7$ **b** $\frac{7m+5}{2}-3=8$ **c** $\frac{4+3x}{4}-5=2$ **d** $\frac{7+2k}{6}+3=12$

e $6+\frac{3w+1}{4}=10$ **f** $4+\frac{6a-5}{3}=9$ **g** $\frac{1-4e}{2}-8=3$ **h** $6+\frac{3-2x}{3}=0$

i $\frac{5-4p}{6}+2=5$ **j** $\frac{7-3y}{4}+1=5$ **k** $\frac{2(x+1)}{5}=10$ **l** $\frac{3(m-2)}{4}=6$

m $\frac{8(n+1)}{3}+2=4$ **n** $\frac{5(1-n)}{2}-1=3$ **o** $\frac{4(1+d)}{3}+1=7\frac{1}{3}$ **p** $8+\frac{3(w-4)}{5}=14$

q $\frac{4(d-5)}{7}-3=4$ **r** $4=\frac{6(3-m)}{5}-4$ **s** $1+\frac{4(7-h)}{3}=13$ **t** $5=2+\frac{3(1-2g)}{7}$

EXAMPLE 4

3 Solve each equation.

a $\frac{2k}{3}=\frac{5}{4}$ **b** $\frac{3w}{10}=\frac{2}{5}$ **c** $\frac{5x}{2}=-\frac{10}{3}$ **d** $\frac{5k-1}{4}=\frac{3}{5}$

e $\frac{5}{4}=\frac{4m+1}{7}$ **f** $\frac{1-2y}{5}=\frac{8}{3}$ **g** $\frac{n-7}{6}=\frac{n+3}{4}$ **h** $\frac{y+2}{5}=\frac{y-1}{2}$

i $\frac{a+5}{3}=\frac{a-1}{8}$ **j** $\frac{p+2}{5}=\frac{p-5}{2}$ **k** $\frac{4k-3}{5}=\frac{2k+7}{4}$ **l** $\frac{3y+2}{3}=\frac{2y+1}{4}$

m $\frac{w}{5}+\frac{w}{2}=7$ **n** $\frac{5a}{7}-\frac{2a}{3}=2$ **o** $\frac{2e}{3}+\frac{e}{2}=8$ **p** $\frac{3a}{2}+\frac{4a}{5}=1$

q $\frac{2y}{5}-\frac{y}{3}=4$ **r** $\frac{a}{3}+\frac{3a}{4}=2$ **s** $\frac{5m}{4}-\frac{3m}{10}=1$ **t** $\frac{4h}{3}+\frac{h}{5}=3$

u $\frac{5y-2}{7}=\frac{3y+5}{3}$

4 Solve each equation. Select **A**, **B**, **C** or **D**.

a $\frac{4m}{5}-\frac{m}{3}=2$

A $m=10$ **B** $m=12$ **C** $m=\frac{30}{7}$ **D** $m=\frac{4}{3}$

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b $\frac{m+1}{2}=\frac{3+2m}{5}$

A $m=1$ **B** $m=5$ **C** $m=\frac{5}{3}$ **D** $m=\frac{2}{3}$

5 Use substitution to match each solution with its equation.

a $x=18$ **b** $x=-1$ **c** $x=3$ **d** $x=4$

A $\frac{3x+1}{2}=5$ **B** $\frac{2x}{3}-\frac{x}{2}=3$ **C** $\frac{x-1}{3}=\frac{x+1}{5}$ **D** $2x-3=x-4$

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Quadratic equations $x^2 + bx + c = 0$ 6.03

An equation in which the highest power of the variable is 2 is called a **quadratic equation**, for example, $x^2 = 5$, $3m^2 + 7 = 10$, $d^2 - d - 6 = 0$ and $4t^2 - 3t = 8$.

The quadratic equation $x^2 = c$

The quadratic equation $x^2 = c$ (where c is a positive number) has 2 solutions:

$x = \pm\sqrt{c}$ (which means $x = \sqrt{c}$ and $x = -\sqrt{c}$).

Example 5

Solve each quadratic equation.

a $m^2 = 16$ **b** $3x^2 = 75$ **c** $4m^2 - 12 = 0$

Video
Simple quadratic equations

Solution

a $m^2 = 16$

$m = \pm\sqrt{16}$ Finding the square root of both sides

$= \pm 4$

b $3x^2 = 75$

$x^2 = \frac{75}{3}$ Dividing both sides by 3

$x^2 = 25$

$x = \pm\sqrt{25}$ Finding the square root of both sides

$= \pm 5$

c $4m^2 - 12 = 0$

$4m^2 - 12 + 12 = 0 + 12$ Adding 12 to both sides

$4m^2 = 12$

$m^2 = \frac{12}{4}$ Dividing both sides by 4

$m^2 = 3$ Finding the square root of both sides

$m = \pm\sqrt{3}$ Leave the answer as a surd (in exact form)

Foundation Standard Complex

Example 6

Solve $7x^2 - 88 = 0$, writing the solution correct to one decimal place.

Solution

$7x^2 - 88 = 0$

$7x^2 = 88$

$x^2 = \frac{88}{7}$

$x = \pm\sqrt{\frac{88}{7}}$

$x = \pm 3.54562...$

$\approx \pm 3.5$

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The quadratic equation $x^2 + bx + c = 0$

To solve quadratic equations of the form $x^2 + bx + c = 0$, factorise $x^2 + bx + c$ into binomial products.



Video
Quadratic equations by factorising

Example 7

Solve $x^2 + 5x + 6 = 0$.

Solution

To factorise $x^2 + 5x + 6$, find 2 numbers that have a sum of 5 and a product of 6.

The correct numbers are 2 and 3.

$x^2 + 5x + 6 = 0$

$(x + 2)(x + 3) = 0$

The LHS has been factorised into 2 binomials, $(x + 2)$ and $(x + 3)$, whose product is 0.

If 2 numbers have a product of 0, then one of the numbers *must* be 0.

$\therefore x + 2 = 0$ or $x + 3 = 0$

$\therefore x = -2$ or $x = -3$

$\therefore$ The solution to $x^2 + 5x + 6 = 0$ is

$x = -2$ or $x = -3$.

Check:

When $x = -2$,

LHS $= (-2)^2 + 5 \times (-2) + 6 = 0$

RHS $= 0$

LHS = RHS.

When $x = -3$,

LHS $= (-3)^2 + 5 \times (-3) + 6 = 0$

RHS $= 0$

LHS = RHS.

Example 8

Solve each quadratic equation.

a $x^2 - x - 2 = 0$ **b** $u^2 + 3u - 28 = 0$

c $3m^2 = 6m$ **d** $w^2 - 10w + 25 = 0$

Solution

a $x^2 - x - 2 = 0$

Find 2 numbers that have a sum of -1 and a product of -2.

They are -2 and 1.

$(x - 2)(x + 1) = 0$

$\therefore x - 2 = 0$ or $x + 1 = 0$

$\therefore x = 2$ or $x = -1$

$\therefore$ The solution to $x^2 - x - 2 = 0$ is

$x = 2$ or $x = -1$.

Check:

When $x = 2$,

LHS $= 2^2 - 2 - 2 = 0$

RHS $= 0$

LHS = RHS.

When $x = -1$,

LHS $= (-1)^2 - (-1) - 2 = 0$

RHS $= 0$

LHS = RHS.

b $u^2 + 3u - 28 = 0$

Find 2 numbers that have a sum of 3 and a product of -28.

They are 7 and -4.

$(u + 7)(u - 4) = 0$

$\therefore u + 7 = 0$ or $u - 4 = 0$

$\therefore u = -7$ or $u = 4$

$\therefore$ The solution to $u^2 + 3u - 28 = 0$ is $u = -7$ or $u = 4$.

c $3m^2 = 6m$

$3m^2 - 6m = 0$

This requires a simpler factorisation as there are only 2 terms, both involving m.

You cannot divide both sides by m because if $m = 0$, then you are dividing by 0.

$3m(m - 2) = 0$

$\therefore 3m = 0$ or $m - 2 = 0$

$\therefore m = 0$ or $m = 2$

$\therefore$ The solution to $3m^2 - 6m = 0$ is $m = 0$ or $m = 2$.

d $w^2 - 10w + 25 = 0$

Find 2 numbers that have a sum of -10 and a product of 25.

They are -5 and -5.

$(w - 5)(w - 5) = 0$

$\therefore w - 5 = 0$ or $w - 5 = 0$

$\therefore w = 5$ or $w = 5$

$\therefore$ The solution to $w^2 - 10w + 25 = 0$ is $w = 5$ (only one solution).

PATH



Video
Quadratic equations by factorising

6.03

PATH

Note: Quadratic equations of the form $ax^2 + bx + c = 0$ can be found in Chapter 14, *Quadratic equations and the parabola*, an optional Advanced chapter.

EXERCISE 6.03 ANSWERS ON P. 596

Quadratic equations $x^2 + bx + c = 0$ UFRC

EXAMPLE 5

1 Solve each quadratic equation, writing the solutions as surds if necessary.

a $m^2 = 144$ **b** $x^2 = 400$ **c** $y^2 = 35$

d $k^2 - 169 = 0$ **e** $y^2 - 1 = 0$ **f** $w^2 - 24 = 0$

g $x^2 + 10 = 14$ **h** $t^2 - 9 = 7$ **i** $\frac{a^2}{2} = 8$

j $4k^2 = 180$ **k** $3w^2 = 300$ **l** $d^2 + 60 = 204$

m $\frac{k^2}{2} = 7$ **n** $\frac{w^2}{10} = 2.5$ **o** $4x^2 = 1$

p $\frac{m^2}{4} = 10$ **q** $5y^2 = 5$ **r** $2p^2 + 3 = 21$

s $\frac{3k^2}{2} + 1 = 13$ **t** $\frac{y^2}{5} - 2 = 18$

EXAMPLE 6

2 Solve each equation, correct to one decimal place where necessary.

a $5m^2 - 20 = 0$ **b** $\frac{4a^2}{9} = 36$ **c** $m^2 = 28$

d $9m^2 - 2 = 32$ **e** $9k^2 + 10 = 13$ **f** $\frac{2x^2}{5} = 23$

g $\frac{k^2}{16} = 6$ **h** $\frac{3k^2}{10} = 27$ **i** $6y^2 = 0.726$

j $3a^2 + 11 = 267$ **k** $2y^2 - 14 = 63$ **l** $\frac{2w^2}{5} - 1 = 19$

3 Solve each quadratic equation. Select the correct answer **A**, **B**, **C** or **D**.

a $x^2 = 121$

A $x = 12, -12$ **B** $x = 11, -11$ **C** $x = 10, 11$ **D** $x = 12, -11$

b $9m^2 - 1 = 35$

A $m = 3, -3$ **B** $m = 2, -2$ **C** $m = 8, -8$ **D** $m = 9, -9$

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4 Solve each equation.

a $x^2 + 3x + 2 = 0$ **b** $y^2 + 5y + 4 = 0$ **c** $y^2 + 16y + 48 = 0$

d $x^2 + x - 12 = 0$ **e** $x^2 + 2x - 3 = 0$ **f** $x^2 + 3x - 40 = 0$

5 Solve each equation.

a $x^2 - x - 30 = 0$ **b** $x^2 - 8x + 16 = 0$ **c** $x^2 - 5x - 66 = 0$

d $d^2 - 2d = 0$ **e** $x^2 - 3x - 10 = 0$ **f** $n^2 + 4n = 0$

g $k^2 - 7k = 0$ **h** $y^2 = 5y$ **i** $v^2 = 12v$

6 Explain why the quadratic equation $x^2 = -25$ has no solutions. R C

7 State which of these quadratic equations have no solutions. Give reasons. R C

a $x^2 = -9$ **b** $2k^2 + 5 = 9$ **c** $3m^2 + 8 = 4$

d $\frac{9w^2}{2} - 1 = 1$ **e** $4 + \frac{d^2}{3} = 8$ **f** $\frac{5a^2}{2} + 3 = 2$

8 Solve each quadratic equation. Select **A**, **B**, **C** or **D**.

a $x^2 + 4x - 60 = 0$

A $x = -10, 6$ **B** $x = 12, -5$ **C** $x = 10, -6$ **D** $x = -12, 5$

b $q^2 + 3q = 0$

A $q = 3, -3$ **B** $q = 6, -3$ **C** $q = 0, -3$ **D** $q = 0, 3$

PATH

EXAMPLE 7

EXAMPLE 8

6.04

Equation problems 6.04

Example 9

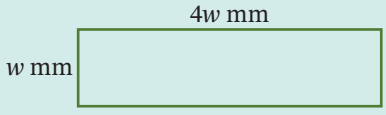
A rectangle is 4 times as long as it is wide. The perimeter of the rectangle is 180 mm. Find the dimensions of the rectangle.

Solution

Let the width of the rectangle be w mm. Then the length is $4w$ mm.

$\therefore$ Perimeter: $w + 4w + w + 4w = 180$

$$10w = 180$$
$$w = \frac{180}{10}$$
$$= 18$$



$\therefore$ The width of the rectangle is 18 mm.

Its length is $4w = 4 \times 18 = 72$ mm.

Check: Perimeter $= 18 + 72 + 18 + 72 = 180$ mm.

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Example 10

Paris is 7 years older than Amy. 10 years from now, the sum of their ages will be 43.

How old are they now?

Solution

Let x = Amy's age now.

Then Paris' age now = $x + 7$.

Break the information into 'Now' and 'In 10 years time'

	Now	In 10 years time
Amy	x	$x + 10$
Paris	$x + 7$	$x + 7 + 10 = x + 17$

In 10 years time:

Sum of ages: $(x + 10) + (x + 17) = 43$

$$2x + 27 = 43$$

$$2x = 16$$

$$x = 8$$

Amy is 8 years old now.

Paris is $8 + 7 = 15$ years old now.

Check: In 10 years time, the sum of their ages will be $18 + 25 = 43$.

EXERCISE 6.04 ANSWERS ON P. 596

Equation problems U F PS R C

For each question, write an equation and solve it to answer the problem. PS R C

1 The longer sides of an isosceles triangle are twice as long as the shorter side. The perimeter of the triangle is 90 mm. Find the lengths of the sides of the triangle.

EXAMPLE 9

2 The length of a rectangle is 3 times as long as its width. The perimeter of the rectangle is 152 mm. Find its dimensions.

3 The length of a rectangle is 3 more than twice its width. Find the dimensions of the rectangle if its perimeter is 84 cm.

4 The sum of 3 **consecutive** integers is 186. Find the integers.

EXAMPLE 10

5 Tyson is 9 times the age of his daughter, Charlotte. In 5 years, he will be 4 times the age of Charlotte. How old are they now?



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6 When 15 is subtracted from 3 times a certain number, the answer is 63. What is the number?

7 The product of 2 and a number is the same as 12 minus the number. Find the number.

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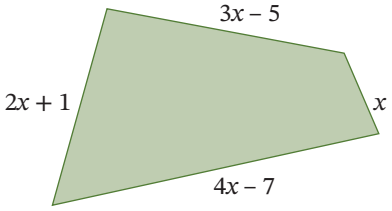
8 The sum of the present ages of Vatha and Chris is 36. In 4 years time, the sum of their ages will equal twice Vatha's present age. How old are they now?

9 4 consecutive integers have a sum of 858. Find the 4 integers.

10 The perimeter of the block of land shown is 344 m.

Measurements shown are in metres.

Find the value of x.



11 Manori's bag contained 10-cent and 20-cent coins. She had 202 coins, with a total value of $31.90. How many 20-cent coins did Manori have?

12 If 17 more than a number is 5 more than 3 times the number, what is the number?

13 The sum of Scott's age and his mother Kait's age is 45. In 5 years time, 3 times Scott's age less 9 will be the same as Kait's age. Find the present ages of Scott and Kait.

14 One angle in a triangle is double the smallest angle, and the third angle in the triangle is 5 more than 4 times the smallest angle. Find the size of each angle in the triangle.

Mental skills 6: Maths without calculators ANSWERS ON P. 596

Quiz
Mental skills 6

Multiplying and dividing by 5, 15, 25 and 50

It is easier to multiply or divide a number by 10 than by 5. So whenever we multiply or divide a number by 5, we can double the 5 (to make 10) and then adjust the first number.

1 Study each example.

a **To multiply by 5**, halve the number, then multiply by 10.

$$\begin{aligned} 18 \times 5 &= 18 \times \frac{1}{2} \times 10 \text{ (or } 9 \times 2 \times 10) \\ &= 9 \times 10 \\ &= 90 \end{aligned}$$

b **To multiply by 50**, halve the number, then multiply by 100.

$$\begin{aligned} 26 \times 50 &= 26 \times \frac{1}{2} \times 100 \text{ (or } 13 \times 2 \times 100) \\ &= 13 \times 100 \\ &= 1300 \end{aligned}$$

c **To multiply by 25**, quarter the number, then multiply by 100.

$$\begin{aligned} 44 \times 25 &= 44 \times \frac{1}{4} \times 100 \text{ (or } 11 \times 4 \times 25) \\ &= 11 \times 100 \\ &= 1100 \end{aligned}$$

d **To multiply by 15**, halve the number, then multiply by 30.

$$\begin{aligned} 8 \times 15 &= 8 \times \frac{1}{2} \times 30 \text{ (or } 4 \times 2 \times 15) \\ &= 4 \times 30 \\ &= 120 \end{aligned}$$

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e **To divide by 5**, divide by 10 and double the answer. We do this because there are 2 5s in every 10.

$140 \div 5 = 140 \div 10 \times 2$

$= 14 \times 2$

$= 28$

f **To divide by 50**, divide by 100 and double the answer. This is because there are 2 50s in every 100.

$400 \div 50 = 400 \div 100 \times 2$

$= 4 \times 2$

$= 8$

g **To divide by 25**, divide by 100 and multiply the answer by 4. This is because there are 4 25s in every 100.

$600 \div 25 = 600 \div 100 \times 4$

$= 6 \times 4$

$= 24$

h **To divide by 15**, divide by 30 and double the answer. This is because there are 2 15s in every 30.

$240 \div 15 = 240 \div 30 \times 2$

$= 8 \times 2$

$= 16$

2 Now evaluate each expression.

a	32×5	**b**	14×5	**c**	48×5	**d**	18×50
e	52×50	**f**	36×25	**g**	28×5	**h**	12×25
i	12×15	**j**	22×35	**k**	$90 \div 5$	**l**	$170 \div 5$
m	$230 \div 5$	**n**	$1300 \div 50$	**o**	$900 \div 50$	**p**	$300 \div 25$
q	$1000 \div 25$	**r**	$360 \div 45$	**s**	$210 \div 15$	**t**	$360 \div 15$

6.05 Equations and formulas

PATH

Puzzle
Getting the right formula

A **formula** is an equation that describes a relationship between variables. For example, the formula for the perimeter of a rectangle is $P = 2(l + w)$, where P is the perimeter, l is the rectangle's length and w is the width. Because the formula is for the perimeter, P is called the **subject** of the formula and it is the variable on its own on the left side of the '=' sign.

Example 11

The cost of a trip, C, in dollars, for a ride share company is $C = 5 + 2.4d$, where d is the distance travelled, in kilometres.

a Find the cost of a trip if the distance travelled is 15 km.

b Find the distance travelled if the cost of the trip was $78.20.



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Solution

a When $d = 15$:

$C = 5 + 2.4 \times 15$

$= 41$

The cost was $41.

b When $C = 78.20$:

$78.20 = 5 + 2.4d$

$73.20 = 2.4d$

$d = \frac{73.20}{2.4}$

$= 30.5$

The distance travelled was 30.5 km.

Example 12

The surface area of a sphere is given by the formula $A = 4\pi r^2$, where r is the radius.

Find (correct to one decimal place):

a the surface area of a sphere with radius 2.8 cm

b the radius of a sphere with surface area 40 m^2.

Solution

a When $r = 2.8$:

$A = 4 \times \pi \times 2.8^2$

$= 98.520...$

≈ 98.5

The surface area of the sphere is 98.5 cm^2.

b When $A = 40$:

$40 = 4\pi r^2$

$4\pi r^2 = 40$

$r^2 = \frac{40}{4\pi}$

$= 3.183...$

$r = \sqrt{3.183}$ $r > 0$ because the radius is positive

$= 1.784...$

≈ 1.8

The radius of the sphere is 1.8 m.

PATH

Videos
Equations and formulas 2
Equations and formulas 1
Formulas and equations

6.05

Video
Equations and formulas 2

PATH

EXERCISE 6.05 ANSWERS ON P. 596

Equations and formulas U F R C

EXAMPLE 11

1 The formula for the perimeter of a rectangle is $P = 2(l + w)$. R

a Find the perimeter of a rectangle with length 10 cm and width 16 cm.

b Find the width of a rectangle whose perimeter is 58 m and length is 12 m.

2 A formula for converting speed expressed in m/s (metres/second) to a speed expressed in km/h is $k = 3.6M$, where M is the speed in m/s. Convert each speed to km/h. R C

a 10 m/s **b** 24 m/s **c** 50 m/s

3 A car is travelling at a speed of 110 km/h on a freeway. Use the formula from question **2** to calculate how fast this is in m/s.

4 The formula for converting temperatures in °F to °C is $C = \frac{5}{9}(F - 32)$. Express each temperature in °C, correct to the nearest degree.

a 80°F **b** 32°F **c** 212°F **d** 102°F

5 Use the formula from Question 4 to express each temperature in °F.

a 30°C **b** 40°C **c** 15°C **d** 45°C

6 The average of 2 numbers, m and n, is $A = \frac{m+n}{2}$. If 2 numbers have an average of 28 and one of the numbers is 13, use the formula to find the other number. R

EXAMPLE 12

7 Pythagoras' theorem is $c^2 = a^2 + b^2$, where c is the length of the hypotenuse in a right-angled triangle and a and b are the lengths of the other 2 sides. Find, correct to one decimal place where necessary:

a c, if $a = 5$ and $b = 10$ **b** a, if $c = 41$ and $b = 40$ **c** b, if $c = 20$ and $a = 10$.

8 The formula for the circumference of a circle is $C = 2\pi r$, where r is the radius. Find, correct to one decimal place: R C

a the circumference of a circle with radius 2.4 m

b the radius of a circle whose circumference is 200 cm.

9 The body mass index (BMI) of an adult is $B = \frac{M}{h^2}$, where M is the mass in kilograms and h is the height in metres. Find, correct to one decimal place: R

a the BMI of Leonie who is 1.85 m tall and has a mass of 72 kg

b the mass of Dane with a BMI of 24, who is 2.1 m tall.

10 The volume of a sphere is $V = \frac{4}{3}\pi r^3$, where r is the radius. Find, correct to one decimal place: R

a the volume of a sphere with radius 3.2 cm

b the radius of a sphere with volume 500 m^3.

11 The average speed in km/h of a car is given by the formula $S = \frac{D}{T}$, where D is the distance covered in kilometres and T is the time taken in hours. Find, correct to the nearest whole number: R

a the average speed of a car that takes 4.5 hours to travel a distance of 420 km

b the distance travelled, if a car maintains a speed of 87.2 km/h for 5 hours

c the time taken, if a distance of 650 km is covered at a speed of 91 km/h.

12 The cost, C (in dollars), of a hire car is $C = 75 + 2.5d$, where d is the number of kilometres travelled. Calculate: R

a the cost of hiring a car to travel 350 km

b the distance travelled, if the cost is $135.

13 The surface area of a closed cylinder is given by the formula $SA = 2\pi r^2 + 2\pi rh$. Calculate, correct to one decimal place: R

a the surface area of a cylinder with radius 2.1 m and height 3.5 m

b the height of a cylinder with surface area 1255.38 cm^2 and radius 9 cm.

PATH

Changing the subject of a formula 6.06

In the formula $v = u + at$, v is the **subject** of the formula. When the formula is rearranged so that one of the other variables becomes the subject, the process is called **changing the subject of the formula**. To change the subject of a formula, use the same rules as for solving an equation. The answer is not a number but an algebraic expression (another formula).

PATH



Video Changing the subject of a formula



Puzzle Formulas squaresaw

Worksheet Changing the subject of a formula

Example 13

a For the formula $v = u + at$, change the subject to a.

b Make x the subject of the formula $d = 3xy - k$.

c If $m = \frac{3d - w}{p}$, change the subject of the formula to w.

Solution

a To make a the subject of the formula, solve it like an equation for a.

$v = u + at$

$u + at = v$ Swapping sides so the new subject a appears on the LHS

$at = v - u$ Subtracting u from both sides

$a = \frac{v - u}{t}$ Dividing both sides by t

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b

$d = 3xy - k$

$3xy - k = d$ Swapping sides

$3xy = d + k$ Adding k to both sides

$x = \frac{d+k}{3y}$ Dividing both sides by $3y$

c

$m = \frac{3d-w}{p}$

$mp = 3d - w$ Multiplying both sides by p

$mp + w = 3d$ Adding w to both sides

$w = 3d - mp$ Subtracting mp from both sides

EXERCISE 6.06 ANSWERS ON P. 596

Changing the subject of the formula U F R C

EXAMPLE 13

1 Make y the subject of each formula. R C

a $3x + y = 5$ **b** $x + 4y = 7$ **c** $d - 6y = 10$

d $g + by = x$ **e** $A = M - xy$ **f** $m = \frac{3y}{d}$

g $\frac{2+wy}{5} = x$ **h** $T = 10gy$ **i** $\frac{y-p}{2k} = d$

j $m = 4h - ay$ **k** $\frac{B-ay}{g} = 6$ **l** $V = \frac{3e+4y}{2}$

m $R = \frac{1}{3}Ty$ **n** $B = \frac{4}{wy}$ **o** $h = \frac{6}{y-1}$

2 For the formula $S = ut + \frac{1}{2}at^2$, change the subject to:

a u **b** a

3 The formula for the area of a trapezium is $A = \frac{1}{2}h(a+b)$. Which of the following is the correct formula for h? Select the correct answer **A**, **B**, **C** or **D**. R C

A $h = \frac{A}{2(a+b)}$ **B** $h = \frac{2A}{a} + \frac{2B}{b}$ **C** $h = \frac{2A}{a+b}$ **D** $h = \frac{2A}{2a+b}$

4 What is the correct formula for $P = 2(L + W)$ if L is the subject? Select **A**, **B**, **C** or **D**. R C

A $L = P - 2W$ **B** $L = \frac{P-2W}{2}$ **C** $L = \frac{P}{2} - 2W$ **D** $L = \frac{P}{2W}$

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5 The cost ($\$C$) of a taxi trip is $C = 3.6 + 2.2d$, where d is the distance travelled in kilometres. R C

a Make d the subject of the formula.

b If the cost of a taxi trip was \$31.10, find the number of kilometres travelled.

6 Change the subject of each formula to the variable shown in brackets. R C

a $y = mx + c \ [m]$ **b** $A = lbh \ [b]$ **c** $V = \frac{1}{3}Ah \ [h]$

d $V = \frac{\pi r^2 h}{3} \ [h]$ **e** $3x + k = 4 - 2x \ [x]$ **f** $P = \frac{d+n}{h} \ [n]$

6.06

Power plus ANSWERS ON P. 597

1 Solve each equation.

a $\frac{3(1-y)}{5} = 4 - 2y$ **b** $\frac{50}{2y} = 10$ **c** $\frac{2m+5}{3} - 1 = m + 4$

2 If $y = \frac{ab + cd}{e}$, find d if $y = -12$, $a = -1$, $b = -8$, $c = 7$ and $e = 4$.

3 Randall is 10 years older than Tarni. In 3 years' time, Randall will be twice as old as Tarni. How old are Randall and Tarni now?

4 One-third of a number added to one-sixth of a number is 18. What is the number?

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CHAPTER 6 REVIEW

Quiz
Language of maths 6

Puzzle
Equations crossword

Language of maths

brackets	check	equation
expand	factorise	formula
fraction	inverse operation	LHS
lowest common multiple (LCM)	quadratic equation	RHS
solution	solve	square root
subject	substitute	variable

1 What type of equation has 2 as the highest power of x?
Write an example of this type of equation.

2 Write the subject of the formula $C = \frac{5}{9}(F - 32)$.

3 Find the lowest common multiple of 6 and 9.

4 Why is it possible for a **quadratic equation** to have more than one solution?

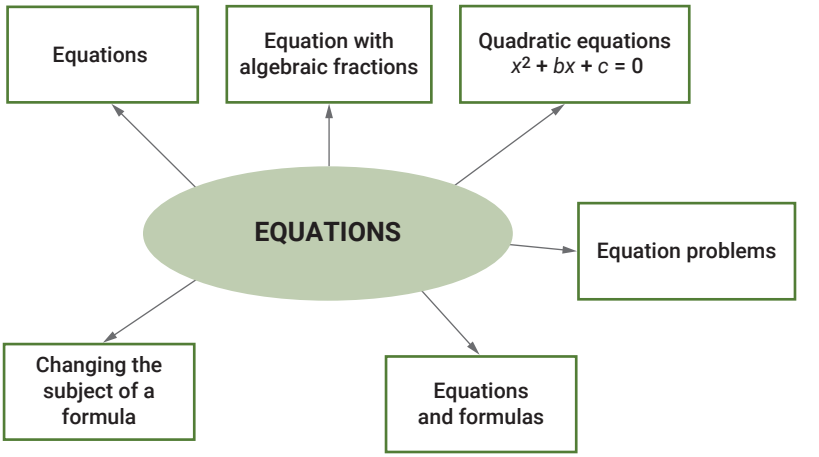
5 When checking the solution to an equation, you need to show that 'LHS = RHS'.
What does that mean?

6 What is the inverse operation to squaring a number?

Worksheet
Mind map: Equations

Topic summary

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.



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TEST YOURSELF 6

ANSWERS ON P. 597

Quiz
Test yourself 6

6.01

1 Solve each equation.

a $3a + 10 = 43$ **b** $8y + 5 = 2y + 21$

c $2a - 12 = 6a$ **d** $9 - 2y = 5 + 2y$

e $3(m - 2) = 27$ **f** $2(2a + 1) = 3(a + 10)$

g $5(h + 1) + 3(h - 2) = 12$ **h** $4(2y + 1) + 3(1 + 4y) = 20$

PATH

6.02

2 Solve $\frac{2(p-1)}{3} = 4$. Select the correct answer **A**, **B**, **C** or **D**.

A $p = 7$ **B** $p = 11$ **C** $p = 5$ **D** $p = 4$

6.02

3 Solve each equation.

a $\frac{3w+2}{5} = 4$ **b** $\frac{y}{5} = \frac{7}{4}$ **c** $\frac{2a+1}{2} = \frac{3a-1}{4}$

d $\frac{3m+5}{6} = \frac{10-m}{3}$ **e** $\frac{4x-7}{3} = \frac{2x+1}{8}$ **f** $\frac{5-7p}{5} = \frac{4p+3}{2}$

g $\frac{2s}{3} - \frac{s}{6} = 2$ **h** $\frac{x}{10} + \frac{x}{2} = 1$ **i** $\frac{5k}{4} - \frac{2k}{5} = 7$

6.02

4 Solve $\frac{4d}{3} + \frac{2d}{5} = 7$. Select the correct answer **A**, **B**, **C** or **D**.

A $d = 9\frac{1}{3}$ **B** $d = 4$ **C** $d = 4\frac{1}{26}$ **D** $d = 8$

6.03

5 Solve each quadratic equation.

a $y^2 = 4$ **b** $p^2 - 100 = 0$ **c** $4x^2 = 40$

d $3m^2 - 3 = 0$ **e** $\frac{2w^2}{5} = 20$ **f** $x^2 + 8x + 7 = 0$

g $h^2 - 8h - 9 = 0$ **h** $u^2 + 4u - 77 = 0$ **i** $k^2 + 5k = 0$

j $m^2 - 2m = 0$ **k** $b^2 + 20b + 100 = 0$ **l** $w^2 = 9w$

6.04

6 The sum of 4 consecutive numbers is 374. Find the 4 numbers.

6.04

7 Corinna is 3 times older than her daughter, Isla. Five years ago, she was 4 times Isla's age. What are their ages?

TEST YOURSELF 6

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6.05

8 The braking distance (in metres) of a bicycle travelling at a speed of v metres/second is $d = \frac{v(v+1)}{2}$. Calculate the braking distance when the speed of the bicycle is 15 m/s.

9 The volume of a pyramid is given by the formula $V = \frac{1}{3}Ah$, where A is the area of the base and h is the perpendicular height of the pyramid. Find:

a the volume of a pyramid with a base area of 48 mm^2 and a perpendicular height of 10 mm

b the base area of a pyramid with a volume of 500 m^3 and a perpendicular height of 5 m.

PATH

6.06

10 Make a the subject of each formula.

a $y = ax + b$ **b** $P = \sqrt{\frac{a}{m}}$ **c** $M(1 + a) = 1 - a$

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